



Dissolved oxygen dependence of Microbe-Derived dissolved organic nitrogen dynamics during membrane biofilm-based linear alkylbenzene sulfonate mineralization

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ABSTRACT

Oxygen-based membrane biofilm reactor (O₂-MBfR) can achieve bubble-free efficient linear alkylbenzene sulfonate (LAS) mineralization. However, effluent microbially derived dissolved organic nitrogen (mDON) reduces recycled water quality and induces water eutrophication. How dissolved oxygen (DO) impacts the dynamics of mDON during LAS biodegradation is unknown. Here, we explored the fate of mDON releasing, molecular characteristics and formation metabolisms in the O₂-MBfR responding to DO dynamics using Fourier transform ion cyclotron resonance mass spectrometry and metagenomic analyses. Results showed that DO presents a negative impact ($\lambda = -0.242$, $P < 0.001$) on mDON production, but can positively influence mDON reduction through influencing total organic nitrogen (TDN) removal. DO of 0.3 mg/L led to the lowest concentration of mDON with the least number of organic molecular formulas and high proportion of stable N₂- and N₃-containing organics. Much lower (0.0 mg/L) or higher (0.6 mg/L) DO concentrations favored dissimilatory-based ammonia formation and assimilatory nitrogen fixation that enhancing mDON formation. Firmocutes were the dominant bacteria that contributing to mDON formation. TDN ($\lambda = 0.844$, $P < 0.001$) and pH ($\lambda = 0.248$, $P < 0.01$) directly affected the effluent mDON concentration. The relatively low effluent mDON under anoxic condition suggested that DO controlling can be an effective strategy to minimize bioavailable mDON discharges in the O₂-MBfR, which showed great implication potential for high-quality reclaimed water production and eutrophication control.

1. Introduction

Total dissolved nitrogen (TDN) in the effluent of wastewater treatment plants (WWPTs) mainly includes dissolved inorganic nitrogen (DIN) and dissolved organic nitrogen (DON) [1]. DON has become a major component (25 %~95 %) of effluent TDN in consequence of the efficient removal of inorganic nitrogen (>95 %) in the WWPTs [2,3]. Particularly, microbially derived dissolved organic nitrogen (mDON) is a principal component of effluent DON [3]. Liquid chlorine is commonly applied for disinfection and residual chlorine can react with nitrogenous compounds including amino acid substances to produce carcinogenic nitrogenous disinfection byproducts [4]. Additionally, mDON can stimulate aquatic algae growth due to its bioavailable properties [3]. Thus, inhibiting mDON production is of practical importance to reduce the headstream generation of chlorinated disinfection by-products and

the risk of water eutrophication.

Since complete eliminating of mDON is impossible, extensive studies were focused on revealing how environmental factors influence mDON production and reduction during wastewater treatment [1,5]. The mDON released from both conventional activated sludge (CAS) systems and pre-denitrification biological nutrient removal (BNR) systems has been shown to stimulate greater phytoplankton production in estuarine waters [6], but mDON in the effluent of BNR system showed much higher bioavailability compared with that of from the CAS system [7]. Moreover, low temperature showed positive effect on unstable mDON production in CAS systems [5]. As a widely used typical N source, ammonia nitrogen produced less mDON than urea in the CAS system [8]. The longer the solids residence time, the lower of DON biodegradability of in the effluent [2]. For green plants and microalgae, chlorophyll *a* (Chl-*a*) in photosystem II (PS II) can cause water oxidation for

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oxygen production [9]. In the ocean water surface (upper 50 m), a positive correlation was found between Chl-a content and DON concentration [10], suggesting that Chl-a may influence mDON metabolism through impacting oxygen production. In the four-stage Bardenpho bioreactor, anoxic zone produced more mDON than the aerobic zone [11].

As a widely used anionic surfactant, linear alkylbenzene sulfonate (LAS) is the dominant organics in the household source-diverted greywater [12,13]. In various biological wastewater treatment systems, LAS has been found to impacts N metabolism [14–16]. The heterotrophic nitrifying-aerobic denitrifying (HN-AD) *Rhodococcus* sp. SY24 could achieve high nitrification and denitrification ratios and excellent N removal capability under low LAS stress [14]. LAS will inhibit anammox biomass activity [16] and affects aerobic denitrification efficiency by influencing related functional enzymes including nitrite reductase [15]. The oxygen-based membrane biofilm reactor (O₂-MBfR) can achieve efficient LAS mineralization with high O₂ transfer rate and non-foaming problems [17–20]. Unidirectional gas supply or air lumen pressure control favored the formation of aerobic-anoxic-anaerobic multifunctional biofilm, which enabled synchronous and efficient removal of both LAS and nitrogen in the O₂-MBfR [21,22]. Despite the wide application of O₂-MBfR in efficient LAS mineralization and nitrogen reduction during greywater treatment [17,19], little is known about how O₂ dosage impacts mDON formation, dynamics and mechanisms during efficient LAS biodegradation in the system, which may influence the effluent quality of O₂-MBfR.

In this work, three O₂-MBfRs with different DO concentrations were applied for LAS treatment. Changes of effluent mDON concentration and composition responding to DO concentration were studied. Fourier transform ion cyclotron resonance mass spectrometry (FTICR-MS) combined with three dimensional excitation-emission matrices (3D-EEM) fluorescence spectroscopy was applied for mDON characterization. The dynamics of microbes and enzymes functional for mDON metabolisms were also tracked, and we related reactor operation parameters and performances to mDON production. The findings enabled low-energy-input and efficient LAS removal and DO controlling-based mDON reduction in the O₂-MBfR.

2. Methods and materials

2.1. Chemicals and LAS-containing wastewater preparation

All the used chemicals, which were of analytical grades, were obtained from Sinopharm Chemical Reagent Co.Ltd., China. The composition of the synthetic LAS-containing wastewater was (in mM) 2.5 KH₂PO₄, 2.5 K₂HPO₄, 0.007 CaCl₂·2H₂O, 0.018 MgCl₂, 0.002 MgSO₄·7H₂O, 0.002 FeCl₂·4H₂O, 4 NaHCO₃, 0.308 LAS, 0.124 NaNO₃, 0.15 NH₄Cl plus 1 mL/L metal trace stock solution (Table S1) [23].

2.2. Setup and operation of membrane biofilm reactor

The O₂-MBfR consists of an acrylic cylinder column (effective volume of 517 mL), multiple rubber hoses, and two peristaltic pumps (BT100-2 J, Longer Pump8, China) for mixing and feeding wastewater. Each reactor contains eight hollow fiber membrane bundles with totally 224 composite polyvinylidene fluoride (PVDF) hollow fiber membranes (28 cm in length, 1.2 mm outside diameter, an average pore size < 0.002 μm, and total surface area of 0.135 m²). These fiber bundles were applied for biomass attachment and biofilm growth. The air pump (SEBO-988) was connected with fiber bundles through the rubber hose to supply oxygen for biofilm formation and pollutants removal. The initial air lumen pressure in three membrane modules were 0.145, 0.580 and 0.870 psi, and DO concentrations inside three reactors were maintained at 0.0, 0.3, 0.6 mg/L by adjusting the air pump every day, respectively.

Activated sludge was obtained from a secondary settling tank at the

Townsend Lake Wastewater Treatment Plant, Wuhan, China, which was then inoculated into three reactors with an initial concentration of 1.5 g total suspended solid (TSS)/L. An internal circulation pump was operated for one day to ensure the effective attachment of biomass onto hollow fiber membrane surface. All the reactors were operated continuously for 90 days with the same hydraulic retention time of 12 h.

Liquid samples were collected from the influent and effluent of reactor every two days, which were initially filtered (0.22 μm, JinTeng, China) and then stored in 50-mL polypropylene centrifuge tubes (BKMAM, China) at 4°C. All samples were used for the analyses of LAS and nitrogen removal and effluent mDON characteristics. At the end of the operation, mixed biofilm samples were collected from each reactor and stored at −20°C for genomic DNA extraction and metagenomic analyses.

2.3. mDON molecular fraction

FTICR-MS and 3D-EEM were applied for mDON molecular fraction analysis. For FTICR-MS analysis, information about the pre-treatment and specific solid phase extraction steps is shown in the [Supporting Information Text 1](#). The samples were then analyzed in negative ion mode using a Bruker Solarix FTICR mass spectrometer carrying an electrospray ion source. Detailed information about the analytical parameters of FTICR-MS is shown in the [Supporting Information Text 2](#). The finalized mass spectral data were analyzed focusing on substances having *m/z* ranging between 100 and 800 Da. Compounds containing N atoms in mDON were separated into CHON and CHONS species and Van Krevelen diagrams were utilized to trace potential components. In the van Krevelen diagram, mDON was classified into seven major substances based on the H/C and O/C ratios of the nature organic matter (NOM) [24]. Details of the partition are given in the [Supporting Information Text 3](#).

3D-EEM combined with parallel factor analysis (PARAFAC) modeling was applied for DOM fractions characterization [25]. EEM raw data were obtained in a luminescence spectrometer (RF-5301pc, Shimadzu, Japan). Distilled water was used as a reference solution to eliminate Raman scattering. Fluorescence intensity was normalized to Raman units (R.U) using the integration of the Raman peak at 350 nm (Em). Fluorescence index (FI) [26], humification index (HIX) [27] and freshness index (BIX) [28] calculated according to EEM data were applied to characterize the properties of DOM components. Detailed information and classification of FI, BIX and HIX was shown in [Table S2](#). Parallel factor analysis (PARAFAC) was conducted using MATLAB R 2022a (MathWorks, USA), with the assistance of the DOM Fluor toolbox [29].

2.4. Microbial community and metagenomic analyses

Biofilm samples were collected from the O₂-MBfRs at the end of the experiment and stored at −20 °C. Biofilm genomic DNA was extracted using the SPINeasy DNA kit for soil (MP Biomedicals, California, USA) following the manufacture's protocol. The extracted DNA samples were sent to Personalbiological Technologies (Shanghai, China) for metagenomic analyses and high-throughput sequencing using the Illumina Misea platform. The raw data was analyzed using Personal Gene Cloud and OIIME2 software (<https://view.qiime2.org/>). Metagenomic-assembled genomes (MAGs) were retrieved using a customized data processing pipeline. All genomes including all available bacterial and algal genomes were classified by Kraken2 (v2.0.12) according to the NCBI RefSeq database [3]. MAGs were identified by applying EnrichM (version 0.5.0) [30]. The relative abundance of functional genes was expressed using normalized reads per million reads.

2.5. Analytical methods

DO concentration inside the reactor was regularly monitored everyday using a portable DO matter (Rex DO-957-Q). A portion of the

effluent sample was filtered using a syringe filter tip (0.45 μm , Jin Teng) and the filtrate was applied to determine nitrate-N (NO_3^- -N), (nitrite) NO_2^- -N, TDN and ammonia (NH_4^+ -N) concentrations. Another fraction of effluent was directly used to determine total residual LAS and COD. Standard methods [31] were used to measure COD, TDN and NH_4^+ -N concentrations. LAS was determined using methylene blue spectrophotometric method [17]. Concentrations of NO_3^- -N and NO_2^- -N were separately measured using nitrate and nitrite kits (HACHB, USA) in a benchtop spectrophotometer (DR 3900, HACH, USA).

Because of the absence of mDON in the synthetic LAS-containing wastewater, the produced mDON was obtained according to the following equation [3]:

$$m\text{DON} = \text{TDN} - \text{NO}_3^- - \text{NO}_2^- - \text{NH}_4^+ - \text{N} \quad (1)$$

2.6. Statistical analysis

Raw data analysis FTICR-MS measurements were performed using OriginLab Origin 2022 and Microsoft Excel 2010 software. The indicators (DO, COD, LAS, NH_4^+ -N, NO_3^- -N, NO_2^- -N, and TDN) were tested in triplicate at specified time intervals during the continuous operation of O_2 -MBfR. Results are expressed as the mean and standard deviation (mean $\pm$ SD). Microbial raw data was downloaded and imported to the online software of OIIME2 for data visualization. Structural equation models (SEM) was evaluated using the maximum likelihood differences obtained from the R language data package and with reference to Rosseel's code [32], which was conducted to assess the direct or indirect effects of DO concentrations on reactor performance and nitrogen metabolism (especially mDON).

3. Results and discussion

3.1. LAS and nitrogen removal performance

Fig. 1 shows the dynamics of LAS and nitrate removal ratios, and the effluent concentrations of TDN and mDON. DO concentration at 0.3 mg/L enabled the highest removal ratio of LAS, which should be contributed to the formation of an aerobic-anoxic dynamic biofilm in the O_2 -MBfR [18,33]. Although previous studies have reported that LAS is more readily biodegraded under high DO conditions, DO at 0.3 mg/L led to more diverse microbial community structure that leading to efficient LAS mineralization [34]. Anoxic biofilm-induced denitrification in the MBfR can further improve LAS removal as which can be carbon source and electron donor [19,22]. For nitrate removal (Fig. 1b), the removal ratio of NO_3^- -N was negatively correlated with DO concentration and DO at 0.0 mg/L achieved maximum nitrate removal (95.5 %). Low DO concentration favored the formation of aerobic-anoxic-anaerobic multifunctional biofilm, which enabled efficient nitrate removal through both aerobic and anaerobic denitrification processes [18,35]. Moreover, nitrate can replace oxygen as an electron acceptor during organics biodegradation under low DO conditions, which improved denitrification-related nitrate removal [36]. DO at 0.3 mg/L led to the lowest effluent concentrations of TDN (1.42 mg/L) and mDON (0.51 mg/L). Consistent with previous findings that microorganisms are more likely to produce mDON under aerobic conditions [37]. Anoxic conditions favored protein-like mDON reduction and mDON can be converted to other substances [38]. Notably, the effluent mDON concentration remains highly consistent with LAS removal rate and high LAS removal ratio led to low effluent TDN and mDON. At low DO conditions (0.0 mg/L), inefficient removal of LAS led to relatively high residual LAS in the system, which can impact the metabolism of microorganisms [15] and enhance the solubility of hydrophobic organic compounds (e.g., polycyclic aromatic hydrocarbons) [39], influencing mDON production and

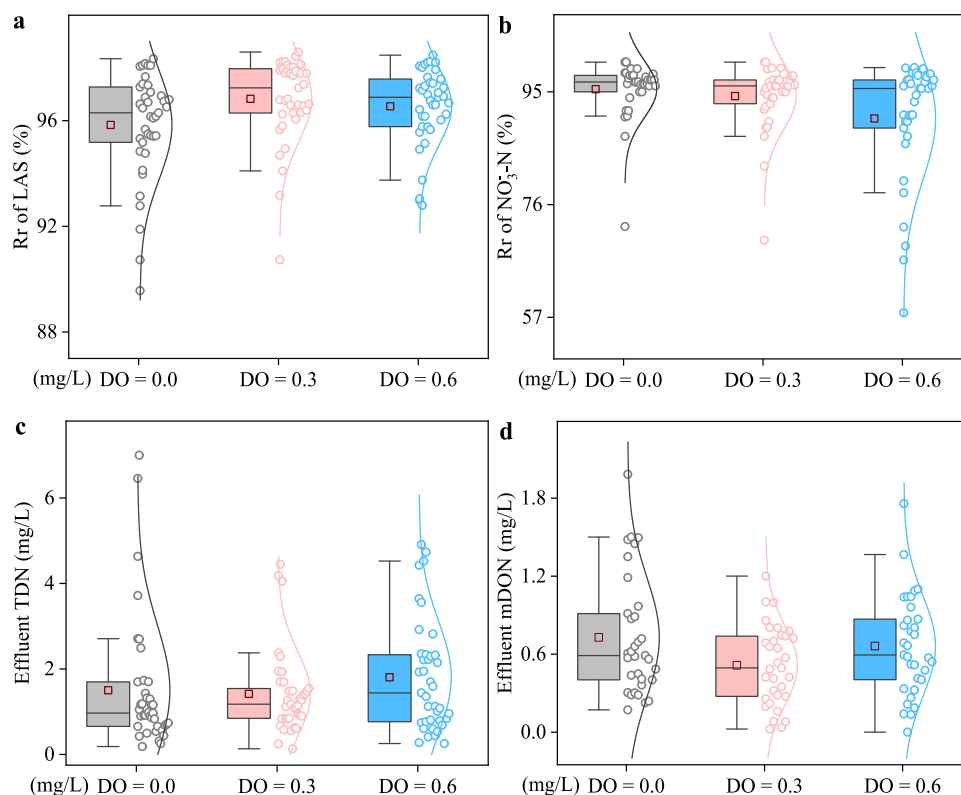


Fig. 1. Dynamics of removal ratios of (a) linear alkylbenzene sulfonate (LAS) and (b) NO_3^- -N, and the effluent concentrations of (c) total dissolved nitrogen (TDN) and (d) microbial-derived dissolved organic nitrogen (mDON) at various dissolved oxygen (DO) concentrations in the oxygen based membrane biofilm reactor (O_2 -MBfR).

solubilization from the biofilm. Moreover, LAS can also bind with biogenic amines [40], leading to the reduced biodegradation and high residual of biogenic amines in the system. All above reasons showed the potential to increase effluent mDON concentration in the system.

3.2. mDON characteristics

Fig. 2 shows the EEM spectra and related EEM-PARAFAC analysis results of the DOM at DO of 0.3 mg/L. Three fluorescence fractions were successfully isolated from the EEM spectra. Extra EEM diagrams are shown in appendix figure S1–2. The three peaks (230 nm/321 nm, 280 nm/321 nm and 395 nm/480 nm) appearing in component1 (C1). The

peaks of 230 nm/321 nm and 280 nm/321 nm represent tyrosine protein [41,42], and peak of 395 nm/480 nm may represents humic acid-like substances [43]. Component 2 (C2) exhibits peaks at 240 nm/299 nm and 290 nm/299 nm representing tyrosine-like proteins [42], and peaks at 240 nm/346 nm and 290 nm/346 nm representing tryptophan-like proteins [44]. In component 3, peaks of 250 nm/429 nm and 315 nm/429 nm were similar to C1 and OC-C1, which were considered as humic acid substances [45,46]. Peaks of 250 nm/321 nm and 310 nm/321 nm are similar to Microbial humic-like substances [46]. The maximum fluorescence intensity of C1 mirrored the changes of mDON that DO of 0.3 mg/L led to the lowest fluorescence intensity of C1. The maximum fluorescence intensity of C2 was negatively correlated with

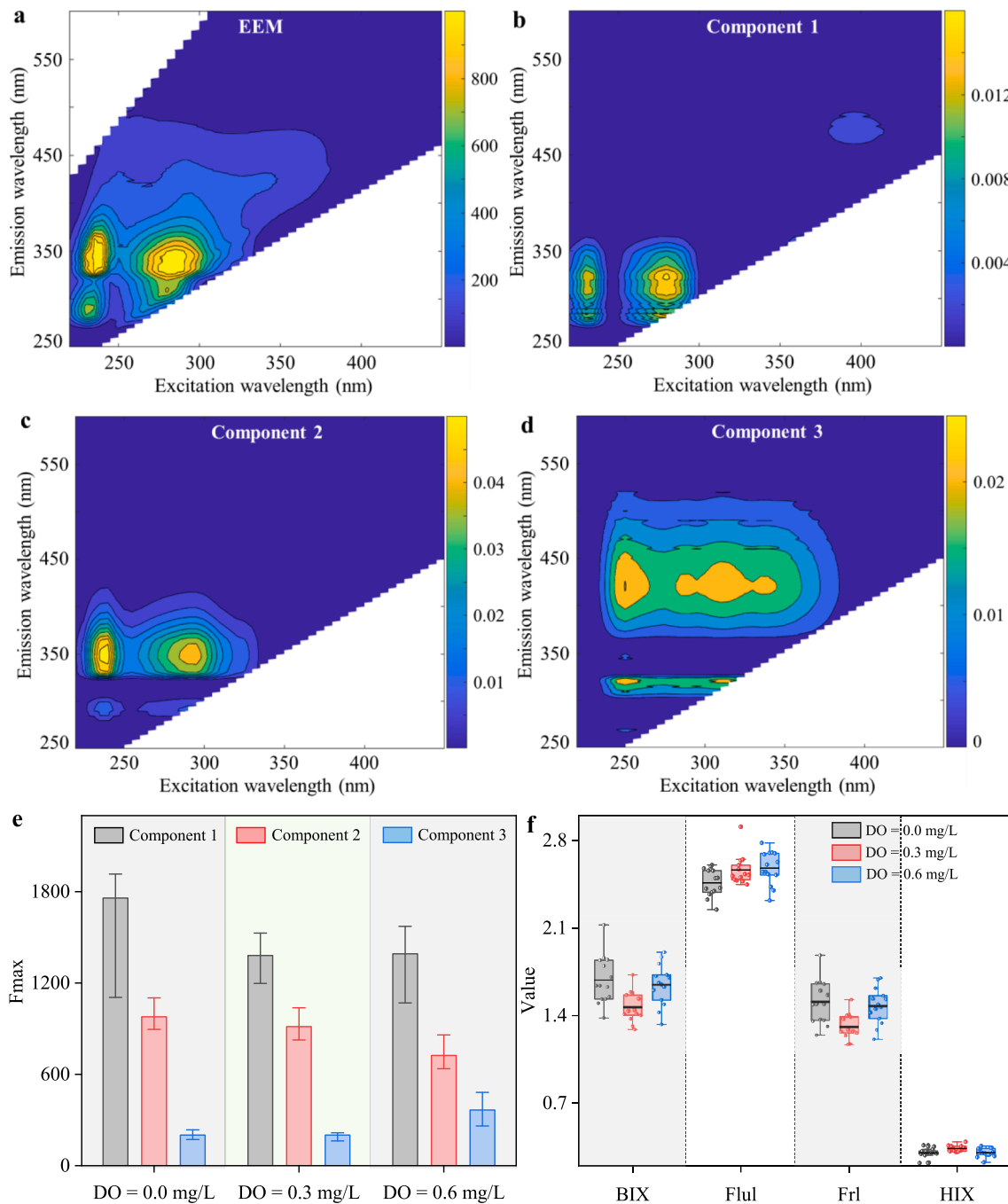


Fig. 2. (a) Selected excitation-emission matrix (EEM) spectra and the related fluorescence components (b) 1, (c) 2 and (d) 3 decomposed using the EEM-PARAFAC analysis, as well as the dynamics of (e) maximum fluorescence intensities of the target components and (f) typical four fluorescence parameters at different dissolved oxygen concentrations.

DO concentration, but the opposite trend for that of C3. Consistent with previous findings that low DO favored tryptophan-like proteins production [42], but high DO favors humic acids production [47]. Generally, low HIX values (<0.7) and high values of FI (>1.9) and BIX (>1.4) at all DO conditions indicated that the fluorescent organics were metabolically produced by the microorganisms with high native microbial activity [28]. The dynamics of BIX values mirrored effluent mDON. BIX index reflects the content of protein-like substances in DOM, and the larger the BIX index, the greater contribution of protein-like fraction in DOM [48]. Freshness index indicates the biological activity of DOM [28]. The lowest values of BIX and FI at DO of 0.3 mg/L indicated the low microorganisms' activity, which may lead to the lowest effluent mDON in the system.

Fig. 3 shows the bubble chart of Van Krevelen diagrams compare nitrogenous and related proportions of nitrogen-containing fractions at different DO concentrations; Fig. 4 shows the related Van Krevelen diagrams based number of formula, proportions of different nitrogen atoms –containing organics and the relative abundance of various nitrogen-containing molecular weights. According to FTICR-MS analysis, the molecules of mDON were divided into eight main categories (carbohydrate-like, A/P-like, lipid-like, UH-like, lignin-like, HOCS-like, AS-like and others) and two subcategories (CHON and CHONS). A clear difference of each substance indicated that DO concentration significantly impacts nitrogen metabolisms in the MBFR. AS-like organics are the dominant fraction in mDON, followed by lignin-like matters. LAS can improve the solubilization of hydrophobic organics containing polycyclic aromatic hydrocarbons [39]. Degradation of aromatic compounds usually relies on mono- or di-oxygenase enzymes initiated under aerobic conditions [42], and high DO concentration favored aromatic compounds biodegradation. DO of 0.0 mg/L led to the highest accumulation of lignin-like mDON, indicating that low DO unfavored the biodegradation of lignin-like mDON [49]. Organic molecules are recognized as unstable substance at $H/C > 1.5$ [1]. Consistent

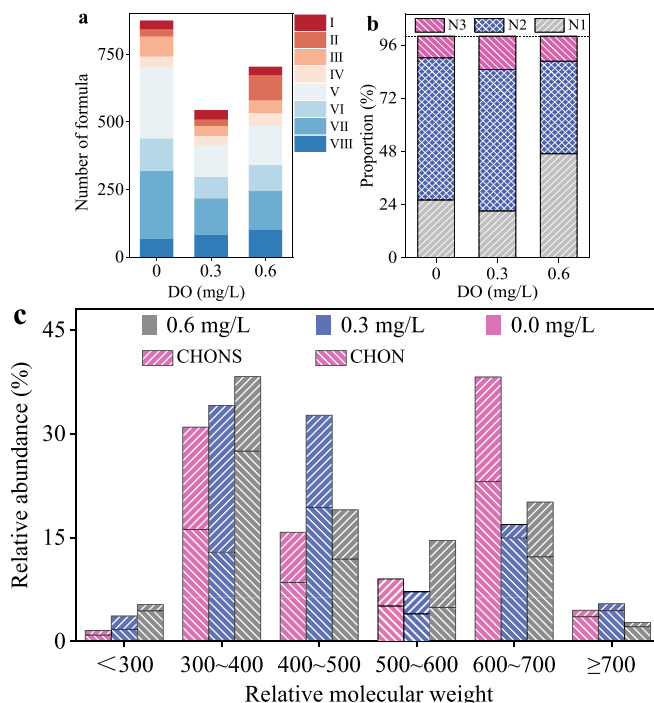


Fig. 4. Van Krevelen diagrams based (a) number of formula, (b) proportions of different nitrogen atoms –containing organics and (c) the relative abundance of various nitrogen-containing molecular weights at different dissolved oxygen concentrations.

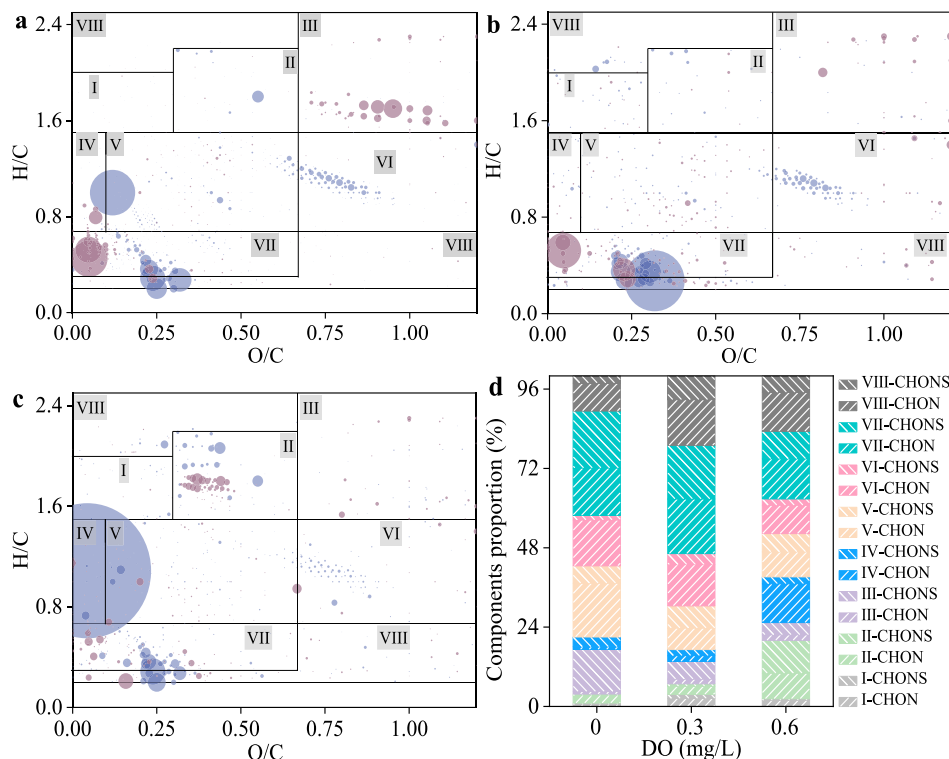


Fig. 3. Bubble chart of Van Krevelen diagrams compare nitrogenous at dissolved oxygen (DO) concentrations of (a) 0 mg/L, (b) 0.3 mg/L, (c) 0.6 mg/L and (d) the related proportions of nitrogen-containing fractions at the noted DO concentrations in the membrane biofilm reactor during linear alkylbenzene sulfonate treatment. Molecules in the diagrams contain (I) lipids, (II) proteins/amino sugars, (III) carbohydrates, (IV) unsaturated hydrocarbons, (V) lignin, (VI) tannins, (VII) condensed aromatics and (VIII) other components.

with BIX value dynamics, the lowest organic molecules (lipids, proteins/ amino sugars, and carbohydrates) with $H/C > 1.5$ at DO of 0.3 mg/L indicated that low microorganisms' activity led to the low produced protein-like mDON [48]. At 0.6 mg/L, Low content and molecular diversity of CHONS at high DO conditions (0.6 mg/L) indicated that high DO can promote complex organics biodegradation and reduce the

complexity of organics molecular structure. DO of 0.3 mg/L led to the lowest diversity of N-containing organic molecules, which should be contributed to the multifunctional biofilm and efficient mineralization of organics under this condition in the O_2 -MBfR, leading to the low diversity of organic molecules in mDON [18,35]. Moreover, low DO led to the accumulation of high N atoms (≥ 2) –containing and high relative

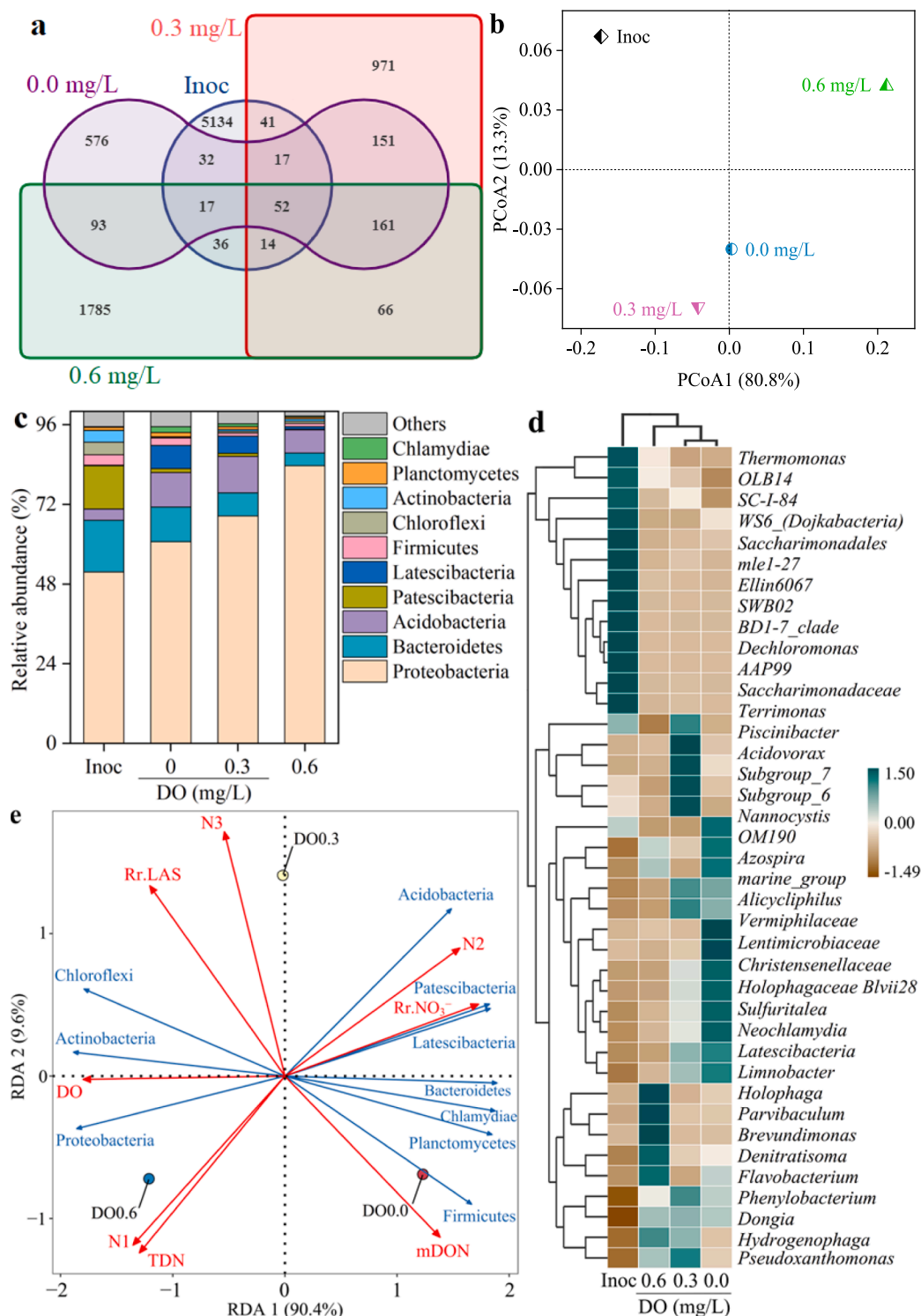


Fig. 5. Dissolved oxygen dependent (a) Number of shared and unique bacterial genera, (b) PCoA of the bacterial communities, microbial communities at (c) phylum and (d) genus levels, and (e) redundancy analysis (RDA) using Bray-Curtis dissimilarity index; Relative abundance of microbial communities at phylum level as response variables (blue arrows); and environmental parameters (containing DO, Rr of LAS & NO₃⁻, effluent TDN and mDON, and different N-containing organics) as explanatory variables (red arrows); Samples from different DO were represented by symbols of different colors. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

abundance of low molecular weights fraction in mDON (Fig. 4b), but high DO concentration favored the accumulation of relatively high molecular weights and more complex molecules in mDON. Results indicated that DO concentration has a great influence on both the accumulation and the molecular composition of mDON produced in MBfR, and that mDON molecules produced under anoxic conditions represent the lowest risk to the environment.

3.3. Synergism of microbes in reducing mDON discharge

Fig. 5 shows the DO-dependent number of shared and unique bacterial genera, PCoA of the bacterial communities, and RDA using Bray-Curtis dissimilarity index. DO concentrations significantly impact microbial communities due to the extremely dispersed clusters of four biofilms samples in the PCoA diagram, which may further impact LAS removal and mDON production in the O₂-MBfR. Compared with the inoculated sludge, the species richness LAS-treated biofilms significantly decreased, which should be resulted from the biotoxicity of LAS [14].

Proteobacteria were the most abundant phylum in all samples and the relative abundance of Proteobacteria also keep increasing when increasing DO concentration, consisting with previous findings that Proteobacteria are functional for organics removal and high DO favored Proteobacteria accumulation and organics mineralization [50,51]. Bacteroidetes and Acidobacteria are other abundant phyla, high DO led to the reduction of Bacteroidetes relative abundance, but DO of 0.3 mg/L achieved the highest abundance of Acidobacteria, which enabled the efficient N reduction. The inoculum-enriched Patescibacteria almost disappeared in the LAS-treated biofilm at high DO conditions, which may contributed to LAS biotoxicity and inappropriate growth under O₂ enriched conditions [51].

At the genus level, the dominant bacteria at DO of 0.0 mg/L were *Limnobacter*, *Latescibacteria*, *Neochlamydia*, *Sulfuritalea*, *Holophagaceae* *Blvii28*, *Christensenellaceae*, *Lentimicrobiaceae*, *Vermiphilaceae*, *OM190*, *Nannocystis* and *Azospira*. Both of *Limnobacter* and *OM190* can degrade aromatic ringed substances including LAS [52,53]. *Christensenellaceae* and *Lentimicrobiaceae* is strictly anaerobic bacteria [54], all of which enabled the efficient organics removal under low DO conditions. The dominant microorganisms at DO of 0.3 mg/L were *Piscinibacter*, *acidovorax*, *Terrimonas*, *Alicyciphilus*, *phenylobacterium* and *Pseudoxanthomonas*. *Pseudoxanthomonas*, *Alicyciphilus* and *Phenylobacterium* were able to degrade aromatic ring-containing substances [55], *Piscinibacter* has been reported to be beneficial for the stabilization of biofilm structures [56], which enabled stable and continuous removal of multiple pollutants. At high DO conditions (0.6 mg/L), *Holophaga*, *Parvibaculum*, *Brevundimonas*, *Denitratisoma*, *Flavobacterium* and *Hydrogenophaga* are the dominant genus. *Brevundimonas* can degrade aliphatic hydrocarbons and polycyclic aromatic hydrocarbons [57], and *Parvibaculum* enabled efficient biodegradation of straight chain alkanes benzene sulfonates [58]. *Holophaga* also produces enzymes that degrade phenolic compounds [59]. Dynamics of DO concentration induced the variations of the dominant denitrifying bacteria, but all of which enabled efficient N reduction. Notably, *Sulfuritalea* that contributing to monosulfur and sulfur trioxide [81] conversion was also detected at DO of 0.0 and 0.3 mg/L, suggesting that sulfur transformation pathway exists in the MBfR at low DO conditions, which may favor organic and N reduction. For RDA analysis, *Firmocutes* had the strongest correlation with mDON, indicating that *Firmocutes* may contribute to mDON production during LAS treatment in the O₂-MBfR. *Firmicutes* can synthesize extracellularly-released proteins by utilizing carbon and nitrogen sources, and these proteins are subsequently hydrolyzed into amino acids and peptides, which are further converted into mDON [60]. The response to different DO concentrations showed a remarkably high correlation between microorganisms and each environmental factor, especially Rr LAS, N3. Similarly, the entire 10 most abundant bacterial phylum, such as *Acidobacteria*, *Planctomycetes*, *Firmocutes*, *Latescibacteria*, *Chloroflexi*, etc., responded significantly to the DO concentration.

This suggests that DO concentration greatly affects the composition of the microbial community in the reactors, which in turn produces different metabolic pathways for organics and nitrogen, leading to the dynamics of pollutants removal and mDON release status.

3.4. Metabolic based strategies in improving mDON reduction

Fig. 6 shows the functional enzymes involved in N metabolism and acquisition during LAS treatment in the O₂-MBfR; Figures S3-S5 present the detailed information about nitrogen assimilation, dissimilation and mineralization related genes, respectively. The N metabolism pathways can be classified into assimilation, dissimilation and mineralization based on metagenomic results. At three DO concentrations, high abundance of glutamine synthetase (*glnA*, *GLUL*), glutamate synthetase (*NADPH*) large chain (*gltB/D*) and nitrate reductase (*NADPH*) (*NR*) were detected in the assimilation pathway. High DO favored the accumulation of *glnA* *GLUL* and *gltB/D*, both of which have been closely linked to glutamate synthesis, which can be further converted to amino acids and newly synthesized biomass [61]. Results suggested that microbes can better utilize ammonium to synthesize new biomass and release surplus mDON at high DO conditions, leading to the high effluent mDON in the system (Fig. 1d). Consisting with previous findings that aerobic condition favored ammonia removal through assimilation pathway rather than nitrification and denitrification pathways during waste leachate treatment [62]. Although high DO concentration (0.6 mg/L) led to relatively high abundance of mDON synthesis-related enzymes in the assimilation pathway, the mDON mineralization-related enzymes (such as nitronate monooxygenase and formamidase) were also enriched in the system. The bacterial genera are more diverse at DO of 0.6 mg/L, which led to more complex interactions between microbial communities, achieving efficient organics mineralization and transformation into newly synthesized biomass. Moreover, high abundance of mDON mineralization-related enzymes and low abundance of mDON synthesis-related enzymes led to the lowest accumulation of effluent mDON at DO of 0.3 mg/L in the system.

Ammoniacal nitrogen plays a key role in mDON synthesis. Four related pathways for ammonia synthesis can be identified in the O₂-MBfR (Fig. 6a) with the functional enzymes of hydroxylamine reductase (*hcp*), hydroxylamine reductase (*nirA*), nitrite reductase [*NAD(P)H*]/nitrite reductase [*NAD(P)H*] small/large subunit (*nasB/D/E*), nitrite reductase (cytochrome *c-552*) (*nrfA/H*) and nitrogenase molybdenum-iron protein alpha chain (*nifD/K/H*). At low DO conditions (0.0 mg/L), dissimilation pathway related ammoniacal nitrogen synthesis enzymes of *nrfA/H* and the enzyme in the assimilation pathway related N fixation enzymes of *nifD/K/H* had the highest abundance. As a key enzyme functional for dissimilatory nitrate reduction to ammonium (DNRA) process, *nrfA* has been applied to quantify microbial populations and functional genes abundance [63]. The bacteria involved in DNRA are mostly facultative and strict anaerobes, and aerobic induced processes coding of *nrfA* gene in bacteria grown can hardly be initiated, which further inhibit DNRA process [64]. Bacteria harboring the *nrfA* gene including *Acidobacteriota*, *Bacteroidota*, *Desulfobacterota*, *Zixibacteria*, *Verrucomicrobiota* and *Myxococcota*, exhibited high abundance at DO of 0.0 mg/L (Figures S4). The *nifD/K/H* enzymes are functional for biological nitrogen fixation (BNF), and low DO is more favorable for the survival of nitrogen fixing bacteria [65]. High abundances of *nrfA* and *nifD/K/H* indicated a more prominent role for DNRA and BNF at low DO conditions, which enhances mDON synthesis of through ammonia nitrogen and creates more conducive mDON formation conditions. Notably, the functional genes related to three nitrogen transformation pathways were mainly found in Actinobacteria and Ascomycetes (Figures S3–5), suggesting that both bacteria play a key role in nitrogen transformation process. Thus, DO concentration regulates the release of mDON by affecting bacteria community structure and enzymes activity functional for nitrogen metabolism, which leads to changes in mDON generation and bioavailability.

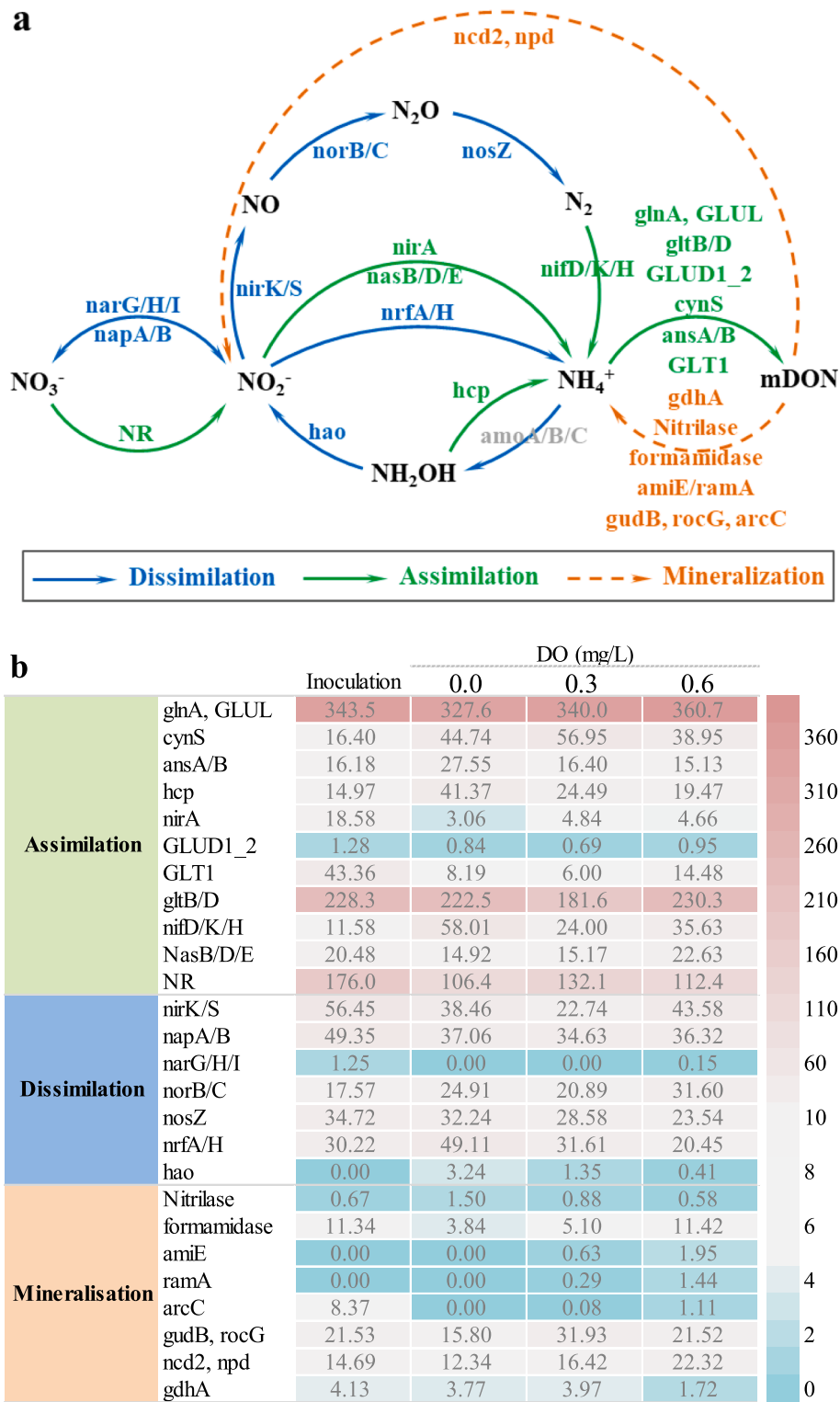


Fig. 6. (a) Proposed nitrogen acquisition and metabolism (according to KEGG pathway map 00910) pathway in the oxygen based membrane biofilm (O₂-MBfR) during linear alkylbenzene sulfonate (LAS) treatment, and (b) the related abundance of functional enzymes in each nitrogen metabolic process. Each colored arrow portraits a nitrogen metabolism process with related functional enzymes. The colors of the bar plots indicate the hosts of corresponding functional genes at the phylum level. The genome-resolved information is available in the Supporting Information.

3.5. SEM revealing the roles of DO in mDON dynamics

Structural equation model (SEM) was performed to quantify the direct and indirect effects of reactor operation effluent physicochemical parameters on mDON production. As shown in Fig. 7, DO concentration

($\lambda = -0.242$, $P < 0.001$), pH value ($\lambda = 0.248$, $P < 0.001$), and TDN ($\lambda = 0.844$, $P < 0.001$) significantly and directly affected effluent mDON concentration. Meanwhile, DO ($\lambda = 0.261$, $P < 0.001$) and NO₂⁻-N ($\lambda = 0.261$, $P < 0.001$) indirectly affect effluent mDON by directly influencing TDN removal, and TDN had the greatest positive impact on

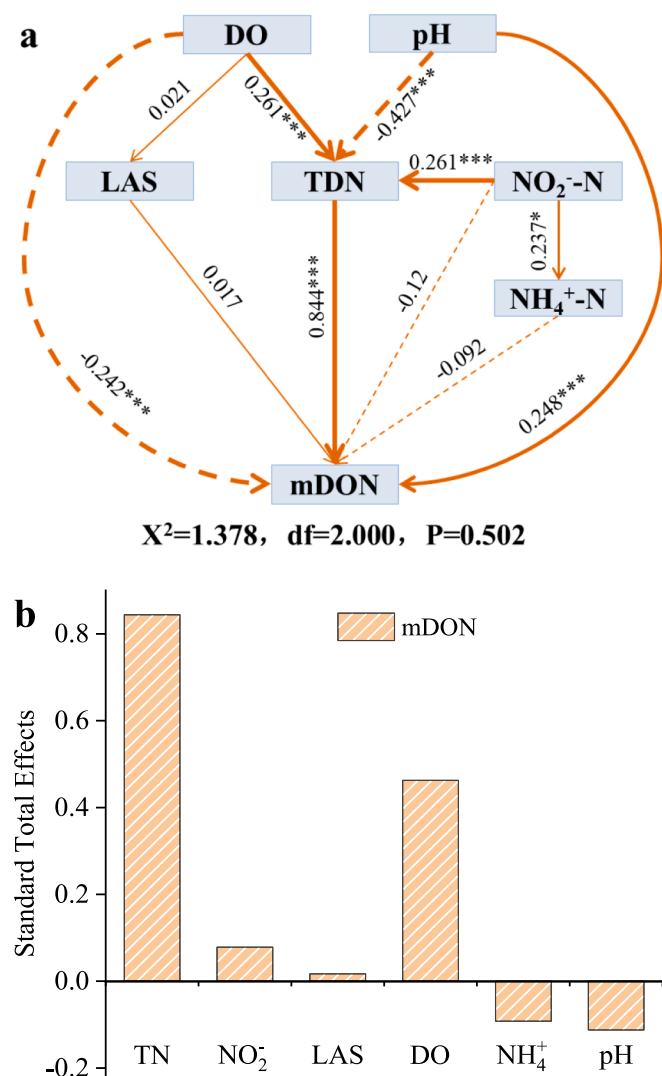


Fig. 7. (a) Structural equation model (SEM) showing the direct and indirect effects of dissolved oxygen concentration on reactor performance (pH, Rr of LAS and NO₃-N, effluent TDN, NO₂-N, NH₄⁺-N and mDON) and (b) the standardized total effects (sum of direct and indirect effects) of typical parameters on effluent mDON discharge. Model $\chi^2 = 1.378$, $P = 0.502$, $df = 2.0$, and $RMSEA = 0.00$. Solid and dotted arrows denote positive and negative pathways, respectively. Numbers near the arrows are the standardized path coefficients, and the width of the arrows is proportional to the strength of the relationships. Asterisks indicate the levels of significance. * $P < 0.05$; ** $P < 0.01$. R² values indicate the proportion of variance explained for each dependent variable.

effluent mDON. Consistent with previous findings that mDON is the dominant component (34 %~49 %) of TDN in the effluent [2,3]. Notably that a positive effect of DO on mDON production was found on the standardized total effect (Fig. 7b), which should be related to that the indirect positive effect of DO on mDON by affecting TDN is more significant than the direct negative effect of DO on mDON generation. Additionally, DO negatively affects mDON production during LAS-containing wastewater treatment in the O₂-MBfR, consistent with previous findings that anoxic conditions favored protein-like mDON biodegradation and mineralization [38]. Results suggested that for the practical application of O₂-MBfR in LAS-containing wastewater treatment, system operation and influent parameters should be regulated according to sewage composition, which can enable the efficient pollutants removal and low emission of effluent mDON.

3.6. Significance and environmental implications

As a widely concerned issue, effluent N after wastewater treatment can cause eutrophication of water bodies. DON can be effectively removed and the increasing proportion of organic N in effluent TN has caused great attention [2,3]. Among effluent TDN, a particular note of interest is mDON, parameters like DO, temperature and hydraulic residence time (HRT) can directly affect the production of mDON [66]. Especially, DO of 0.3 mg/L achieved the lowest concentration of mDON, which also contains less unstable fractions including proteins, lipids, and carbohydrates, and more stable compounds including condensed aromatics (AS), tannins (HOCs), and other substances. The produced mDON also exhibits the lowest number of molecular formulas with low abundance of small molecular weight fractions. Dissimilatory-based (assimilation-based) ammonia formation and assimilatory nitrogen fixation are the main ways in which microorganisms promote mDON synthesis, while microbial mineralization of organic matter is an effective vehicle for microorganisms to reduce mDON accumulation. Overall, O₂-MBfR enabled efficient mineralization of LAS, which suppressed mDON production at DO of 0.3 mg/L. Regulating DO concentration should be an effective strategy that enabling efficient pollutants removal, TDN reduction and alleviating mDON production during LAS treatment in the O₂-MBfR. Although O₂-MBfR system can achieve stable LAS and nitrogen removal under low oxygen conditions when using nitrate as a nitrogen source, the effect of system operation conditions, nitrogen species and environmental stress on mDON production remains unclear. Thus, mDON releasing characteristics and control strategies in O₂-MBfR during sewage treatment should be systematically and deeply studied.

4. Conclusions

During LAS mineralization in the O₂-MBfR, AS-like organics are the predominant mDON, followed by lignin at all DO concentrations. High relative abundance of CHON substances are mainly distributed in the AS-like mDON, which also show higher diversity than CHONS organics. DO concentration of 0.3 mg/L achieved efficient removal of LAS and nitrogen, the lowest concentration of effluent mDON ($H/C > 1.5$, biodegradable), and the minimum number of mDON molecular formula. At high DO conditions (0.6 mg/L), mDON contains higher molecular weight organics, and the organics-biodegradation related Proteobacteria is largely enriched in the system. DO concentration significantly impacts the diversity and relative abundance of functional microorganisms that related to nitrogen metabolism, and Firmicutes phylum was the dominant contributor to mDON formation. DO, pH and TDN directly affected the effluent mDON concentration. Ammonia accumulating from assimilation and dissimilation processes is more conducive to mDON formation at low DO conditions (0.0 mg/L). Thus, maintaining the system at anoxic condition by controlling DO concentration is more favorable for removing TN and reducing mDON formation and release during LAS treatment in the O₂-MBfR.

CRediT authorship contribution statement

Ting Ran: Writing – original draft, Visualization, Software, Methodology, Formal analysis. **Ting Wei:** Writing – original draft, Visualization, Software, Methodology, Formal analysis. **Libo Xia:** Writing – review & editing, Visualization, Software. **Weikang Rong:** Writing – review & editing, Visualization, Methodology. **Yun Zhou:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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